

Project Title

DocsGuide: An Assistant for Government Document Procedures

Project Summary

DocsGuide is an AI-powered chatbot that helps people navigate government document procedures like citizenship certificates, passports, and birth registrations. It provides clear, step-by-step guidance, lists the required documents, and offers conversational support to clarify legal questions and concerns. Through a simple and interactive conversational interface, DocsGuide aims to reduce confusion and save time for citizens dealing with complex paperwork.

Problem Statement

In Nepal, citizens often struggle with navigating government document procedures such as obtaining citizenship certificates, passports, birth registrations etc. due to complex processes, inconsistent information, and limited accessibility to official guidance. This leads to confusion, delays, and multiple visits to government offices. The project seeks to address these challenges by developing an AI chatbot that provides clear, conversational assistance to simplify and streamline these essential document procedures for Nepali citizens.

Objectives

Key objectives that the project aims to achieve:

- To develop an AI chatbot that offers step-by-step guidance and conversational help on government document procedures and related legal queries for Nepali citizens.
- To manually collect, verify, and structure data from official government sources to ensure accurate and reliable information for the chatbot.

Application Areas

DocsGuide is primarily intended to assist citizens in navigating government document procedures. Beyond this, the underlying AI chatbot framework can be adapted for broader applications in public service sectors, including legal aid, administrative support, and other government-related information services where conversational assistance can enhance accessibility and user experience.

Project Justifications

DocsGuide addresses the common challenges Nepali citizens face when dealing with complex government document procedures by providing clear, accessible guidance. It reduces time, effort, and errors in paperwork, improving overall

efficiency. This project promotes digital inclusion and supports the government's move toward more citizen-friendly services through AI-powered assistance.

Limitations and Constraints

- Manual data collection may limit the scope and timeliness of information updates.
- The chatbot's accuracy depends on the quality and completeness of the collected data.
- Language nuances and complex legal terminology might challenge conversational clarity.
- Limited project duration may restrict the number of government procedures covered.

Team Member Details

Name	Email	Github
Andis Paudel	andispaudel@gmail.com	https://github.com/and-is
Dipin Adhikari	adhikari.dipin2@gmail.com	https://github.com/Dipin-Adhikari
Utsab Dahal	utsabdahal34@gmail.com	https://github.com/utsab345
Bikash Pokhrel	thebikashpokhrel@gmail.com	https://github.com/thebikashpokhrel

Requirements

Expected Inputs

- Manually collected, structured data on government documents:
- User queries in natural language(both Nepali or English) during chatbot interaction.

Expected Outputs

- An interactive AI chatbot assisting users in government services navigation
- Clear, conversational guidance on document procedures
- Lists of required documents for various processes
- Answers to legal and procedural questions

User Queries and Expected Chatbot Responses

User Query	Expected Response
नागरिकता प्रमाणपत्रका लागि कुन-कुन कागजात चाहिन्छ?	नागरिकता बनाउनका लागि मुख्यतया यी कागजातहरू चाहिन्छन्: • आमा र बुवा दुवैको नागरिकता प्रमाणपत्र: सक्कल प्रतिलिपि र फोटोकपी दुवै। • जन्म दर्ता प्रमाणपत्र: तपाईंको जन्म दर्ता भएको प्रमाण। • शैक्षिक योग्यताको प्रमाणपत्र: एसएलसी/एसईई वा सोभन्दा माथिको। • फोटो: पासपोर्ट साइजका दुई प्रति फोटो। • बसाइसराइको प्रमाणपत्र: यदि तपाईं बसाई सरेर आउनुभएको हो भने। • विवाह दर्ता प्रमाणपत्र: विवाहित महिलाको हकमा विवाह दर्ता र पतिको नागरिकता। • स्थायी बसोबासको प्रमाण: जग्गाधनी पूर्जा, बिजुली वा पानीको बिल जस्ता घर ठेगाना खुल्ने कागजात। यी सबै कागजातहरू लिएर सम्बन्धित जिल्ला प्रशासन कार्यालय वा इलाका प्रशासन कार्यालयमा जानुपर्छ।
नागरिकता प्रमाणपत्रमा भएको विवरण सच्याउन कसरी सकिन्छ?	यदि नागरिकता प्रमाणपत्रमा उल्लेखित थर, उमेर वा अन्य सानातिना त्रुटी सच्याउनु परेमा, आवश्यक प्रमाणसहित तोकिएको अधिकारीसमक्ष निवेदन दिन सकिन्छ। निवेदन परेपछि सम्बन्धित अधिकारीले प्रमाण र साक्षी समेत बुझी पहिलेको नागरिकता खिचेर अर्को नागरिकताको प्रमाणपत्र दिन सक्छ। विवाह वा सम्बन्धविच्छेद भएकी महिलाले थर वा वतन संशोधन गर्न चाहेमा पनि यही प्रक्रिया अपनाइन्छ।
What are the two types of passport services available in Nepal and where can I apply for each?	There are two types of passport services available in Nepal: normal and express. Normal Service: You can apply at the District Administration Office or Area Administration Office. This service is also available at Nepali missions abroad. Express Service: This service is exclusively available at

	the Department of Passport in Tripureshwor, Kathmandu.
--	---

End User & Stakeholders Requirements

- Ask questions about government document procedures through the chatbot
- Receive step-by-step guidance for completing applications
- Clarify doubts related to legal requirements and documentation
- Use the chatbot as a quick reference to reduce errors and save time during paperwork

Functional Requirements

- The system shall understand and process natural language queries from users.
- The chatbot shall provide step-by-step guidance on government document procedures.
- The chatbot shall answer common legal and procedural questions related to government documents.
- The system shall support bilingual interaction (Nepali and English).

Non Functional Requirements

- **Usability:** The chatbot interface should be simple and intuitive for users with varying technical skills.
- **Accuracy:** Responses must be based on verified and up-to-date government information.
- **Performance:** The system should respond to user queries within a few seconds to ensure smooth interaction.
- **Maintainability:** The system should allow easy updates to the knowledge base as government procedures change.

Data Collection

Documents Covered

- Citizenship Certificate
- Birth Registration Certificate
- Passport

Data Identification

- Textual data on government procedures

- Step-by-step instructions
- Eligibility criteria
- Legal laws and related documents
- Lists of required documents for three target documents

Data Sources

- Citizenship Certificate
 - *Official Websites:* [Ministry of Home Affairs](#), [DAO, Bhaktapur](#), [DAO, Darchula](#)
 - *Publications:* [Citizenship Act 2063\(Nepali\)](#), [Citizenship Act 2063\(English\)](#)
- Birth Registration Certificate
 - *Official Websites:* [Gokarneshwor Municipality Office](#)
 - *Publications:* [Births, Deaths and Other Personal Events \(Registration\) Rules](#)
- Passport
 - *Official Websites:* [Department of Passport](#)
 - *Publications:* [Documents related to Official, Service, and Diplomatic Passports](#), [Passport Regulation, 2020 \(2077\)](#), [Passport Act, 2019 \(2076\)](#)
- Additional verification sources such as [Nepal Law Commission](#), Local Ward Offices

Data Acquisition Methods

- Manual review of official government websites
- Downloading and extracting data from PDFs and published documents
- Cross-checking information across multiple official sources
- Using automated web scraping if allowed and feasible
- Collecting updates periodically to maintain accuracy

Data Collection Challenges

- **Incomplete or outdated info** - cross-check and update regularly
- **Limited online data** - use official publications and manual sources
- **Data accuracy** - rely on verified government sources only
- **Legal limits on scraping** - comply with terms, prefer manual collection
- **Language variations** - standardize terms and support bilingual data

Evaluation Metrics

Retrieval Precision@k (P@k): Fraction of retrieved documents among the top k that are relevant to the query.

Recall@k (R@k): Fraction of all relevant documents retrieved within the top k .

Answer Accuracy (%): Proportion of responses that are factually correct and match official government guidelines.

Information Completeness (%): Percentage of answers that include *all* necessary steps and required documents.

Language Clarity Score (User Rated): How clear and understandable the chatbot's responses are, considering legal and procedural terms.

User Satisfaction Score (1–5 or %): Collected through surveys after interaction.

Bilingual Support Success (%): Percentage of successful queries answered correctly in Nepali and English.